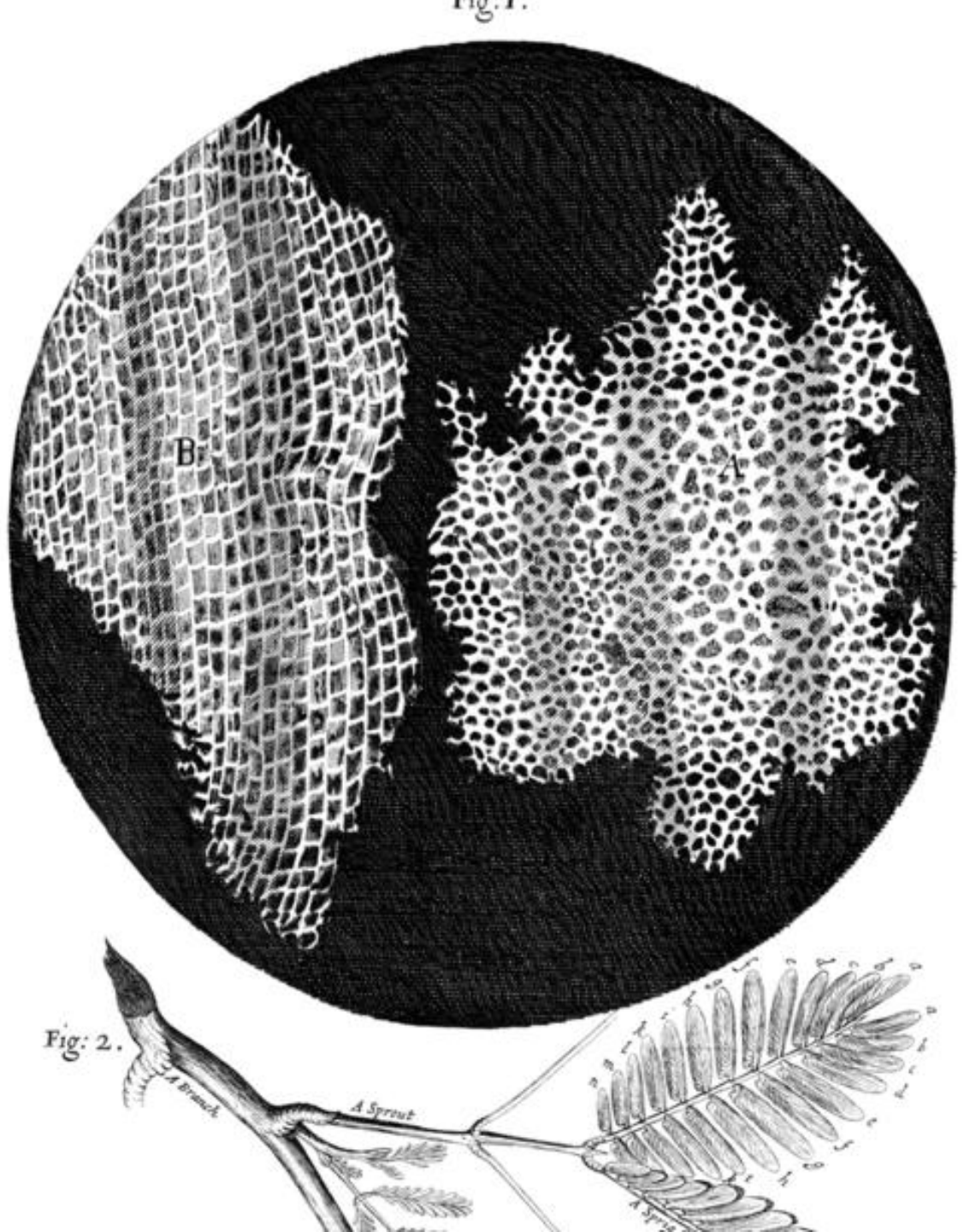
DP Biology – A2.2

# Cell structure

- What do all cells share, and how do they differ?
- How do we use microscopes to study cell structure?

SL statements A2.2.1–11. Textbook pp. 49–81.
Image: cork cells, as drawn by Robert Hooke in 1665. Public domain.

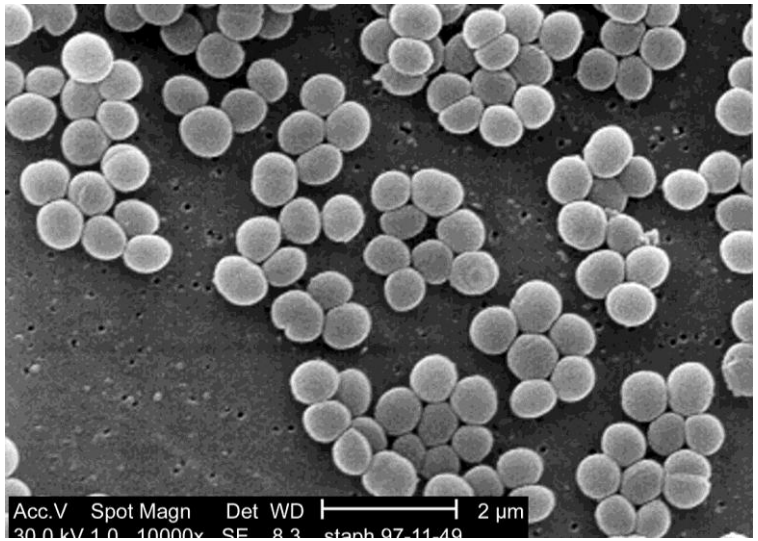


# Objectives for each class

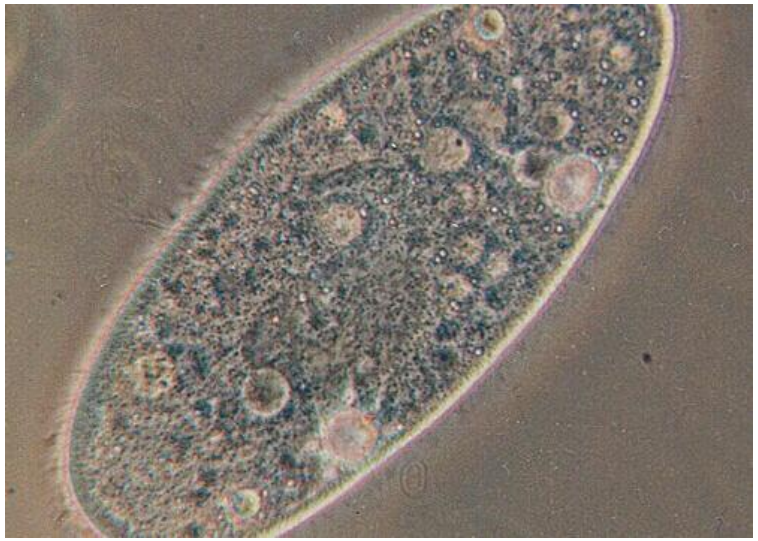
| Class | Statements | By the end of the class, you can... |
|---|---|---|
| **1** | A2.2.1–3 | Explain cell theory, use a microscope, calculate magnification and size, and compare microscopes |
| **2** | A2.2.4–6 | Identify structures in all cells, and compare prokaryotic and eukaryotic cells |
| **3** | A2.2.7–11 | Explain life in one cell, compare animal, fungal and plant cells, atypical cells, and read and draw micrographs |

# From bacteria to you

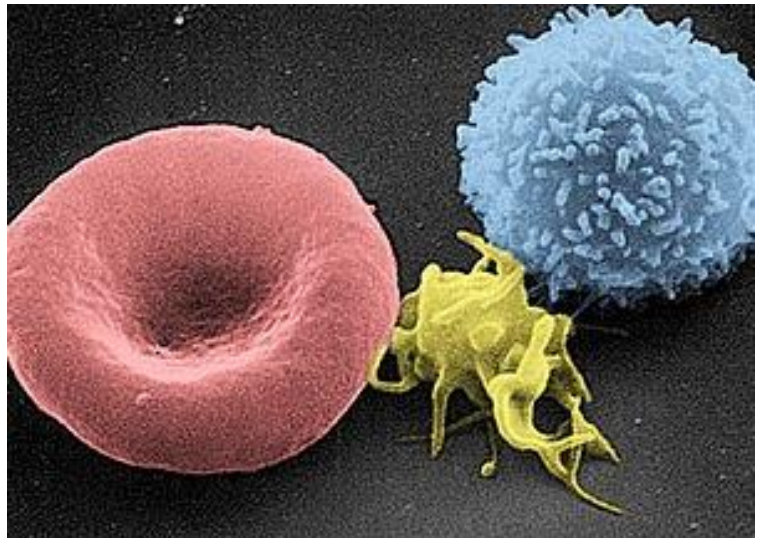
A2.2.1–6




Bacteria (Staphylococcus, SEM). Janice Carr, public domain.



Paramecium. Barfooz, CC BY-SA 3.0.



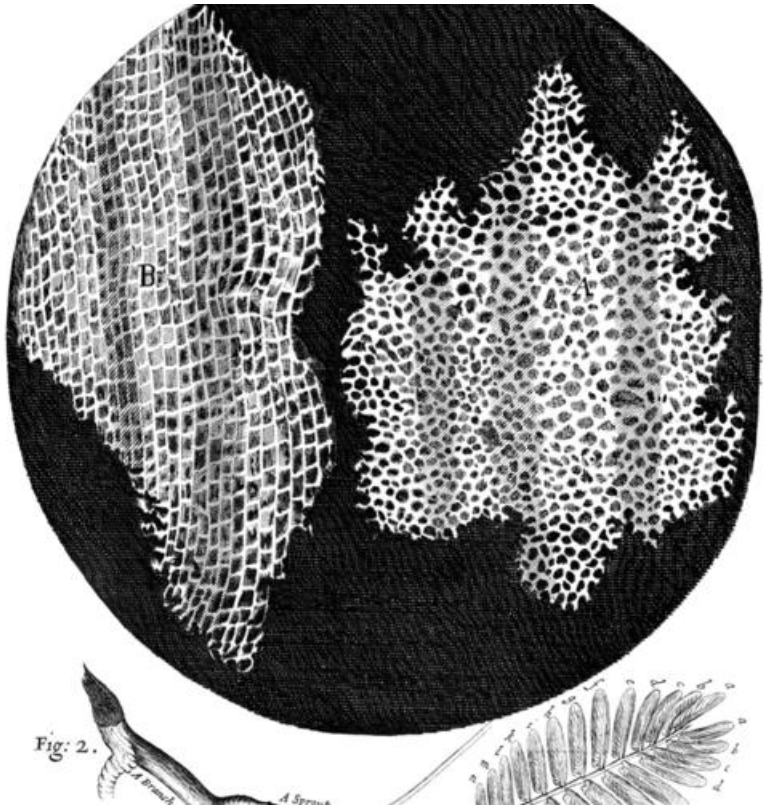
Human blood cells (SEM). NCI, public domain.

A bacterium, a pond organism and your own blood are all made of **cells**. They look very different, yet they share the same basic parts.

This chapter answers two questions: what do all cells have in common, and how do we see something so small?

# Cell theory

A2.2.1




Cork cells drawn by Robert Hooke, 1665. Public domain.

All living things are made of one or more cells

The cell is the smallest unit of life

New cells only come from existing cells

### Using the theory

A theory lets us predict. If a new organism is discovered, cell theory predicts it will be made of cells. Each confirmed prediction makes the theory stronger.

# Using a light microscope

A2.2.2



Compound microscope. Acagastya, CC0.

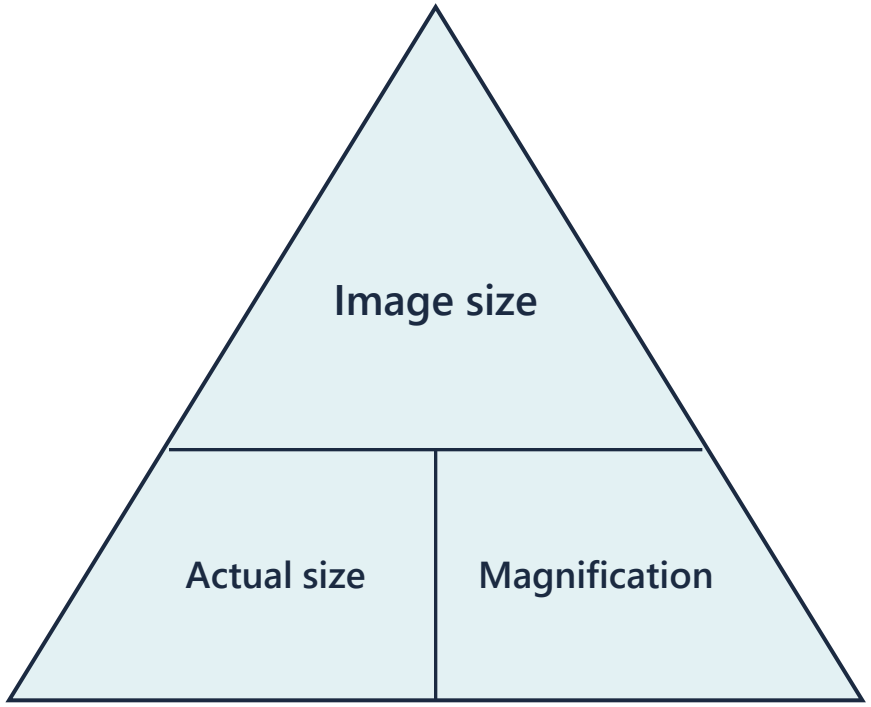
1. Make a thin mount: specimen in a drop of water, coverslip lowered at an angle
2. Stain it (for example methylene blue) to show structures
3. Focus at low power with the coarse knob, then use the fine knob
4. Measure with an eyepiece graticule calibrated against a stage micrometer

**Try it**

Mount onion epidermis, measure one cell with the graticule, and work out its real length in µm.

# Magnification and size






Cover the one you want; the others show the sum.

**Magnification = image size / actual size**




### Golden rule

Convert both sizes to the same unit before dividing. Cells are measured in µm; viruses and ribosomes in nm.

# Worked examples



### Find the magnification

A red blood cell is 8 µm wide. In a drawing it is 40 mm wide.

40 mm = 40 000 µm
40 000 / 8 = × 5000

### Find the actual size

A micrograph at × 2000 shows a cell 50 mm long.

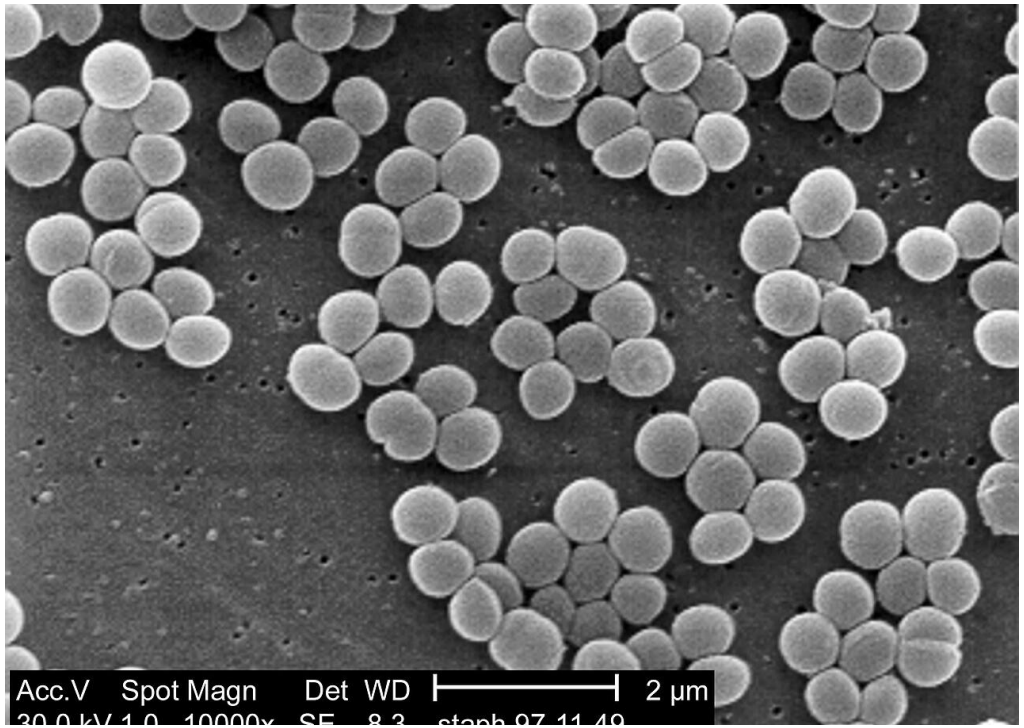
50 mm = 50 000 µm
50 000 / 2000 = 25 µm

### Check your answer

Is it sensible? A cell should be a few to about 100 µm. An answer of 25 000 µm (2.5 cm) means the units went wrong.

# Using a scale bar

A2.2.2




Staphylococcus (SEM). Janice Carr, public domain.

A scale bar shows how long a real distance looks on the image.

Measure the bar with a ruler. Then: magnification = bar length on image / length it represents.

Use that magnification to find the size of anything else in the image.

### Try it

This image has its own 2 µm bar (bottom left). Measure it on your printout, work out the magnification, then find the width of one bacterium. (Staphylococcus cells are about 1 µm across.)

# Light or electron microscope?

A2.2.3

| | Light | Electron |
|---|---|---|
| **Uses** | light | electron beam |
| **Resolution** | about 200 nm | about 1 nm |
| **Living cells?** | yes | no (vacuum) |
| **Colour** | yes | no; added later |



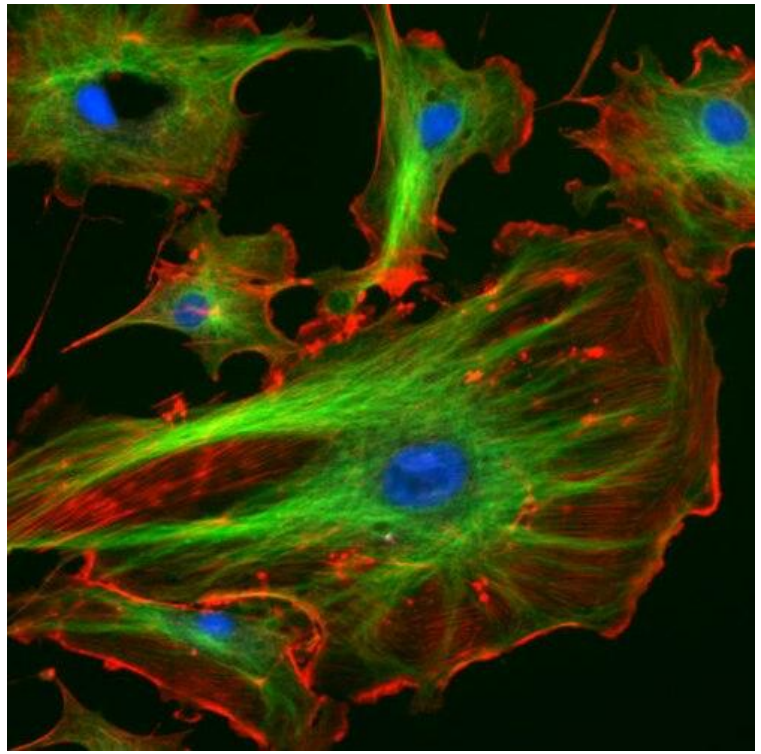
Mitochondria seen with an electron microscope. Louisa Howard, public domain.

### Resolution is what matters

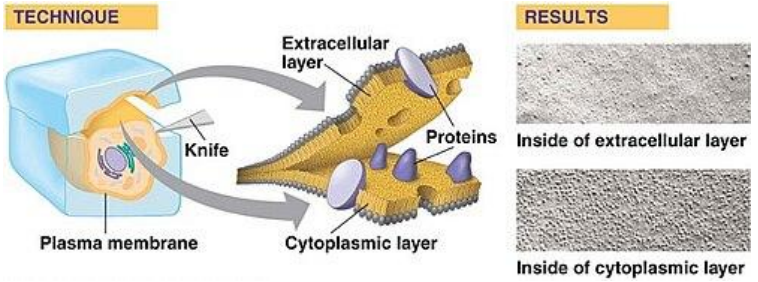
Resolution is the smallest distance at which two points can still be seen as separate. Electrons have a much shorter wavelength than light, so an electron microscope shows far more detail: ribosomes, membranes and the inside of mitochondria.

# New ways of seeing cells

A2.2.3



Fluorescent stains. Public domain.




Freeze fracture. Charles Mallery, public domain.

**In the fluorescence image, the nuclei are blue, the cytoskeleton fibres are green and red.**

| Technique | What it shows |
|---|---|
| **Fluorescent stains** | Chosen structures glow in colour |
| **Immunofluorescence** | Antibodies carry a dye to one target |
| **Freeze fracture** | Splits membranes to show their inside |
| **Cryo-EM** | Frozen molecules seen in their natural 3D shape |

# Practice: size and magnification

A2.2.2

Image sizes are invented for practice; real sizes are typical values.

| Structure | Image size | Magnification | Real size |
|---|---|---|---|
| **Red blood cell** | 35 mm | x 5000 | ? |
| **Mitochondrion** | 40 mm | ? | 2 µm |
| **Bacterium** | ? | x 10 000 | 3 µm |
| **Ribosome** | 5 mm | x 200 000 | ? |

1. Complete the table, with units.
2. Rank the four by real size.
3. Which could you see with a light microscope?

# Check-in: class 1

A2.2.1–3

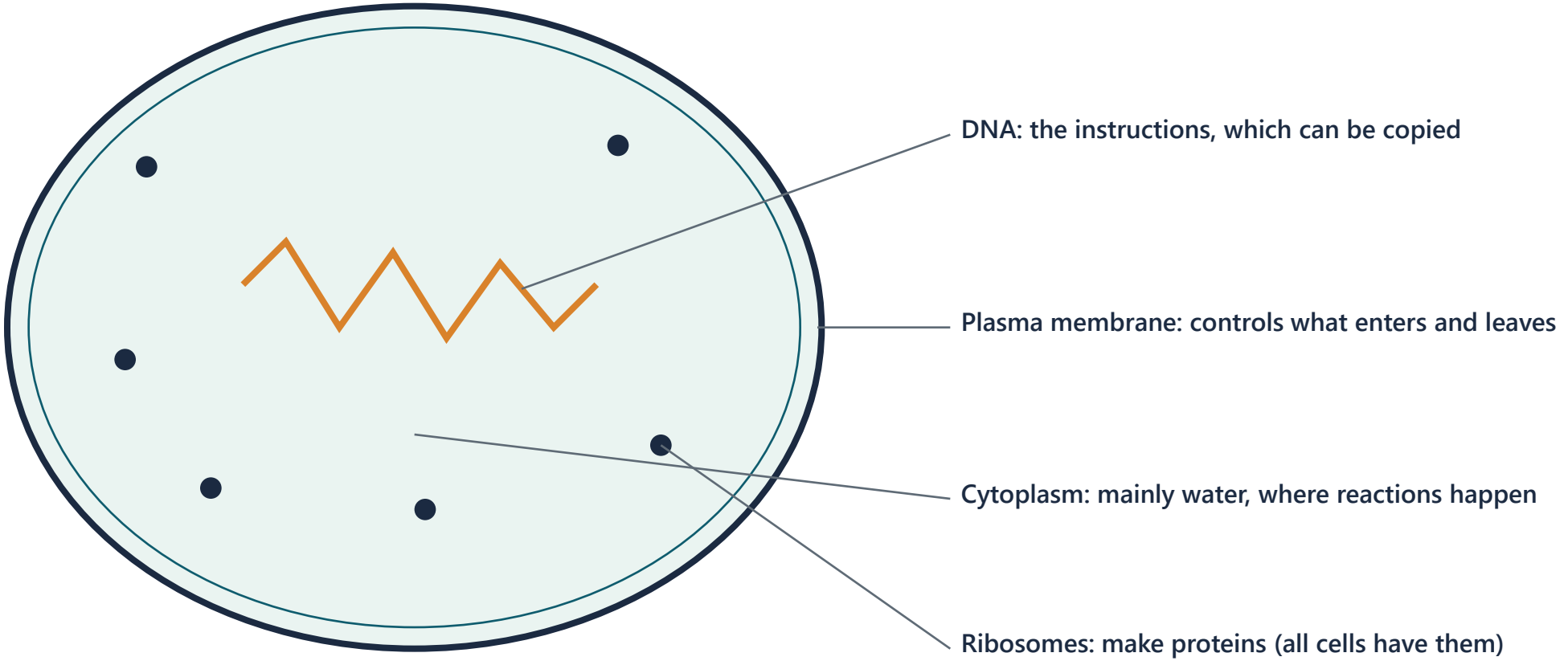
1. State the three parts of cell theory.
2. A cell 20 µm long is 40 mm long in an image. Calculate the magnification.
3. Convert 0.5 mm into micrometres.
4. Explain why an electron microscope shows more detail.
5. State one advantage of a light microscope.
6. Name one newer technique and what it shows.

**How to use this**

Answer without your notes, then check against slides 4–10.

# What every cell has

A2.2.4

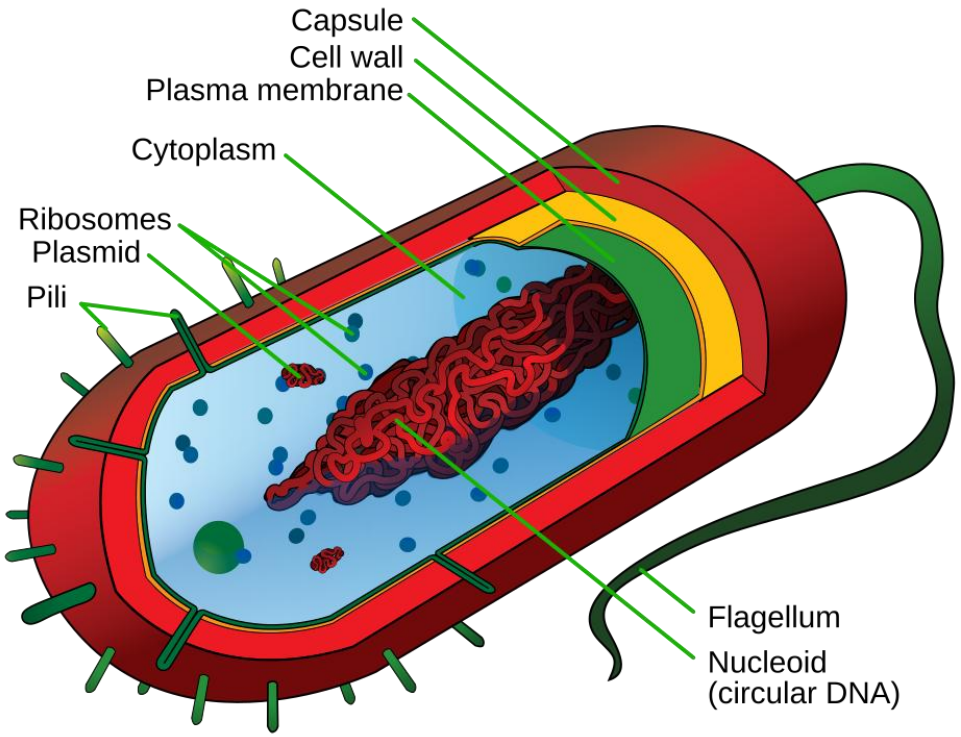



**Why?**

Every cell, from bacterium to neuron, shares these. That fits the idea that all life has a common ancestor.

# Prokaryotic cells

A2.2.5




A generalized prokaryote. LadyofHats, public domain.

**Prokaryotes** (bacteria) have **no nucleus**: their DNA is a **naked loop** in an area called the **nucleoid**.

There are **no membrane-bound organelles**, so the cytoplasm is not divided up.

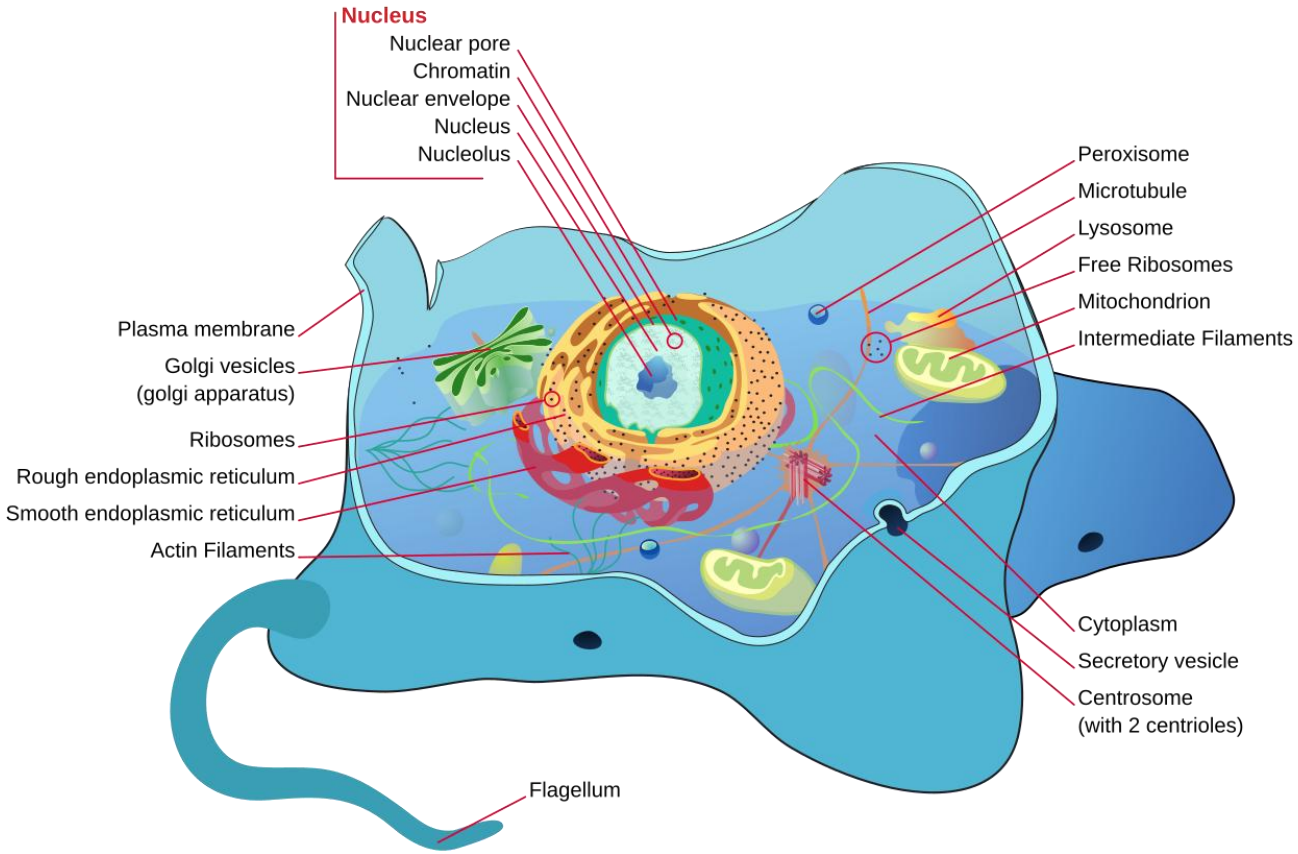
They have a **cell wall**, small **70S ribosomes**, and often **plasmids**, **pili** and a **flagellum**.

### Example to know

Gram-positive bacteria such as Staphylococcus and Bacillus. Be ready to draw one from an electron micrograph.

# Eukaryotic cells

A2.2.6




An animal cell. LadyofHats, public domain.

**Eukaryotic** cells have a **nucleus** surrounded by a double membrane, and DNA wound around proteins (histones) as chromosomes.

The cytoplasm is divided into **compartments**: membrane-bound organelles, each with its own job.

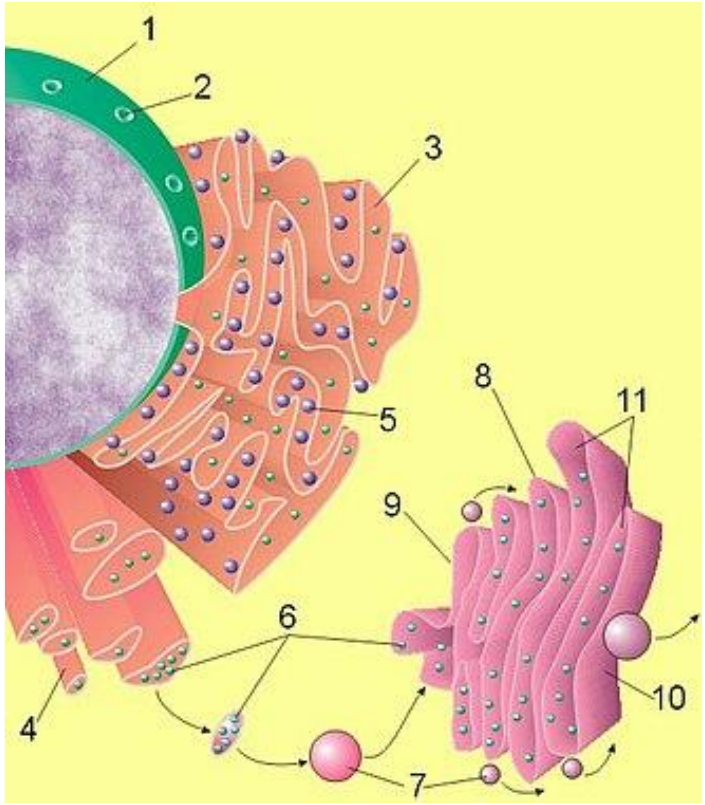
Larger ribosomes (80S) and a **cytoskeleton** of protein fibres.

### Why compartments?

They keep enzymes and substrates together, and keep harmful enzymes (in lysosomes) away from the rest of the cell.

# Organelles 1: the protein factory

A2.2.6




Nucleus, ER and Golgi (diagram). Public domain.

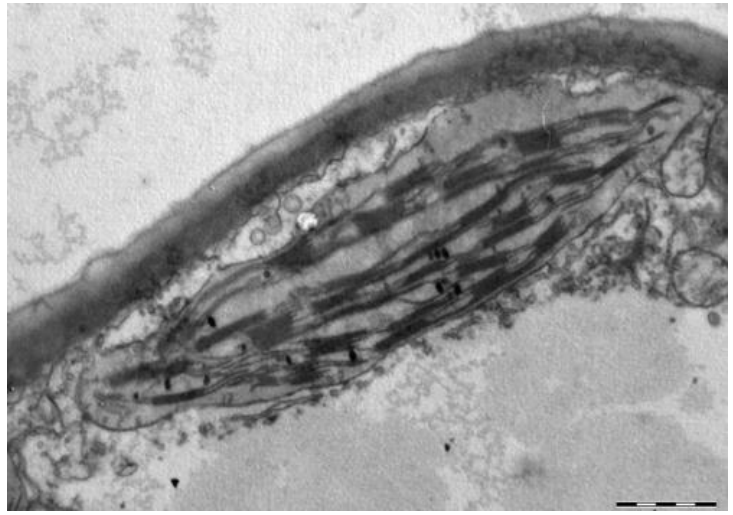
| | |
|---|---|
| **Nucleus** | DNA; transcription |
| ↓ | |
| **Ribosomes** | make polypeptides |
| ↓ | |
| **Rough ER** | makes proteins for export |
| ↓ | |
| **Golgi apparatus** | modifies and packs them |
| ↓ | |
| **Vesicles** | carry them to the membrane |

# Organelles 2: energy and upkeep

A2.2.6



Mitochondria (TEM). Louisa Howard, public domain.



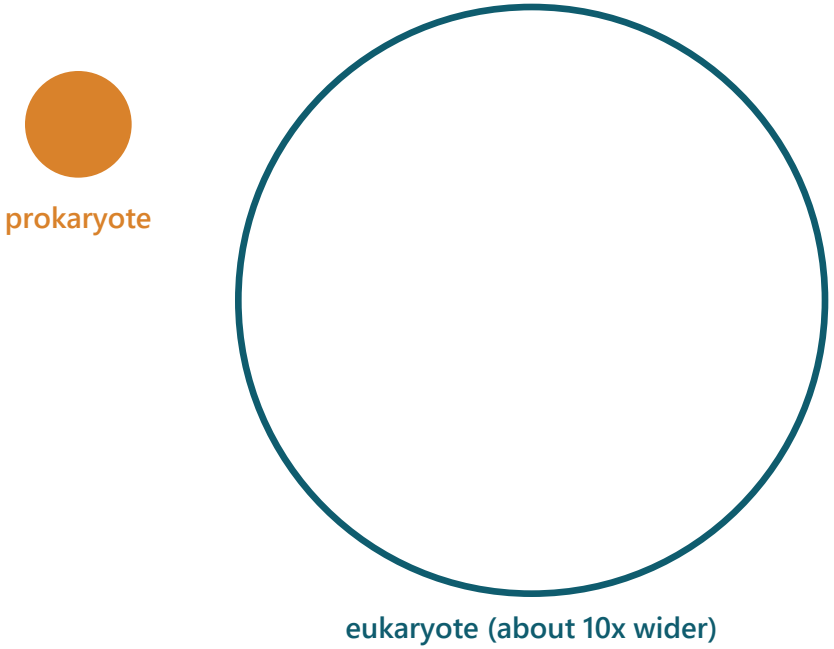
Chloroplast (TEM). and3k and caper437, CC BY-SA 3.0.

| Organelle | Job |
|---|---|
| **Mitochondrion** | Aerobic respiration: makes ATP |
| **Chloroplast (plants)** | Photosynthesis |
| **Lysosome** | Digests old parts and food |
| **Vacuole** | Stores water and solutes |
| **Cytoskeleton** | Shape, movement, moves organelles |
| **Centrioles (animals)** | Organize cell division |

A2.2 Cell structure – Textbook pp. 68–72

# Prokaryotes and eukaryotes compared

A2.2.5–6

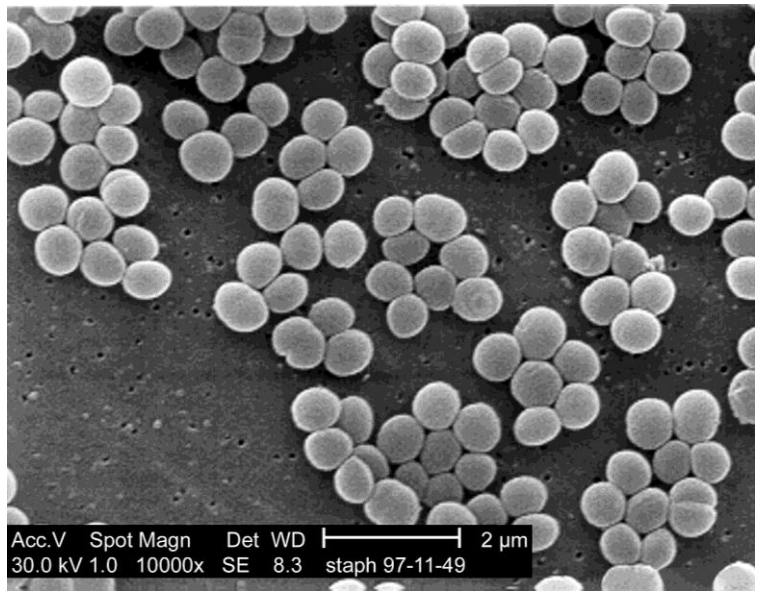



| | Prokaryote | Eukaryote |
|---|---|---|
| **Nucleus** | No | Yes |
| **DNA** | Naked loop | With histones |
| **Organelles** | None | Many |
| **Ribosomes** | 70S | 80S |
| **Size** | 1–5 μm | 10–100 μm |

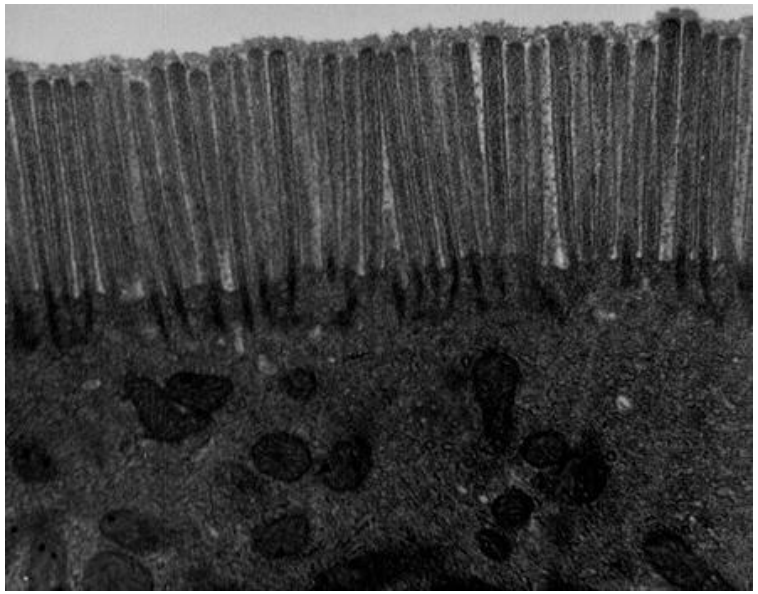
**Both have: DNA, cytoplasm, a plasma membrane and ribosomes.**

# Practice: which kind of cell?

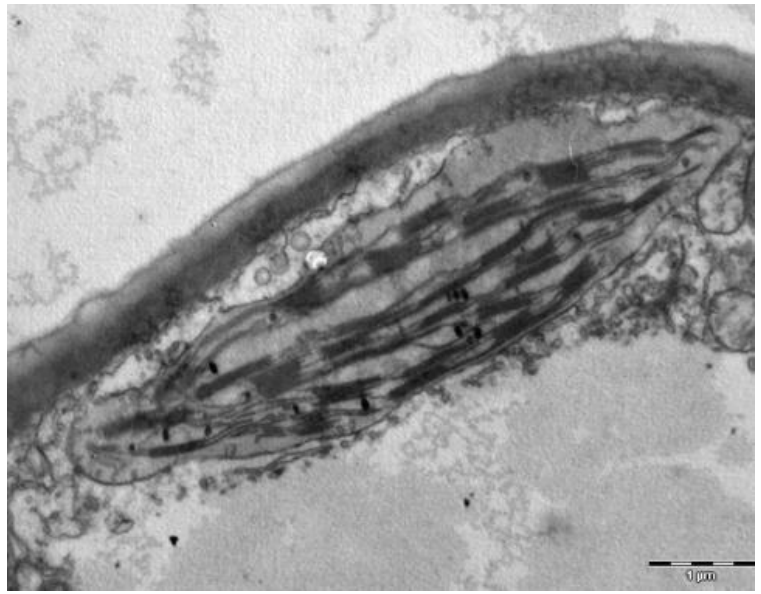
A2.2.10




A



B



C

1. For each micrograph A, B and C, state whether it is a prokaryote, an animal cell or a plant cell.
2. Give the clue you used each time.

# Check-in: class 2

A2.2.4–6

1. State three structures found in all cells.
2. Describe the DNA of a prokaryote.
3. State two structures found only in eukaryotes.
4. Explain one advantage of compartments in a eukaryotic cell.
5. Outline the path of a protein made for export.
6. Compare the ribosomes of prokaryotes and eukaryotes.

**How to use this**

Answer without your notes, then check against slides 13–18.

# Life in a single cell

A2.2.7



Paramecium. Picturepest, CC BY 2.0.

| | |
|---|---|
| **Nutrition** | food vacuoles |
| **Homeostasis** | contractile vacuole |
| **Growth** | takes in materials |
| **Reproduction** | divides in two |
| **Movement** | cilia beat |
| **Excretion** | through the membrane |
| **Response** | swims from harm |
| **Metabolism** | enzymes in cytoplasm |

**Compare**

Chlamydomonas, a single-celled alga, does all the same, but makes its food by photosynthesis.

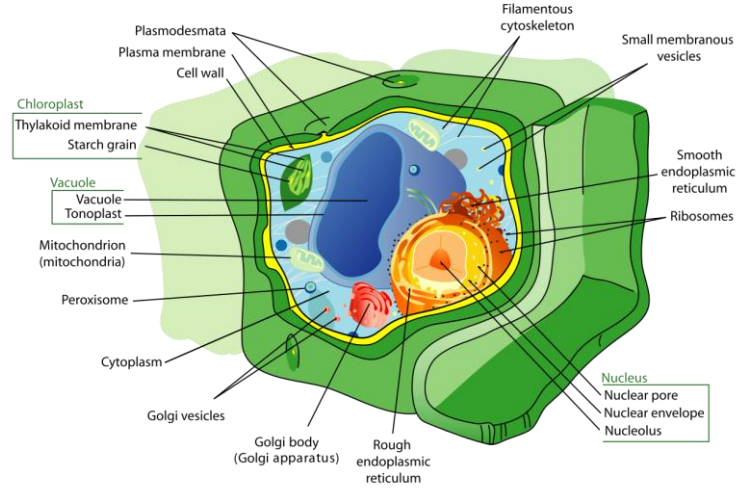
# Animal, fungal and plant cells

A2.2.8

| | Animal | Fungus | Plant |
|---|---|---|---|
| **Cell wall** | none | chitin | cellulose |
| **Chloroplasts** | no | no | yes |
| **Big vacuole** | no | sometimes | yes |
| **Centrioles** | yes | no | no |
| **Cilia/flagella** | often | rarely | rarely |

### Watch out

Not all plant cells have chloroplasts (root cells do not). Fungi are not plants: their walls are chitin, the same as insect skeletons.




A plant cell. LadyofHats, public domain.



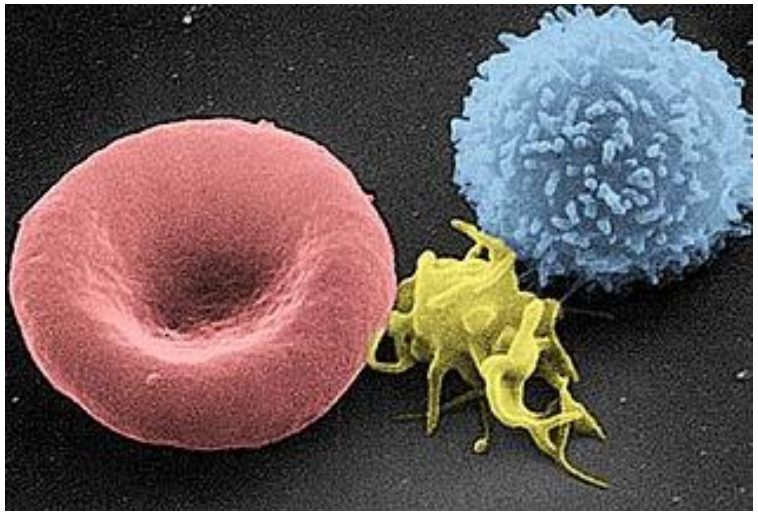
Yeast, a fungus. Murtey and Ramasamy, CC BY 3.0.

# Cells that break the rules

A2.2.9



Skeletal muscle fibres. CC BY-SA 3.0.



Red blood cells. NCI, public domain.



Fungal hyphae. SCIdude, CC0.

### Skeletal muscle

Many nuclei: cells fused to make one long fibre that contracts as a unit.

### Red blood cells

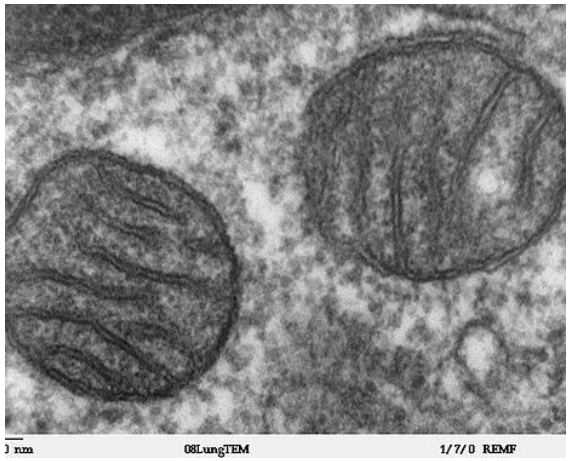
No nucleus: more room for haemoglobin, and more flexible. Phloem sieve tubes also lose theirs.

### Aseptate hyphae

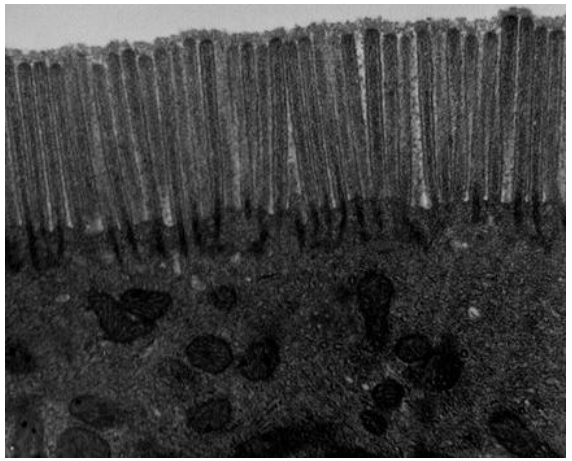
Many nuclei in one long tube with no cross-walls, sharing one cytoplasm.

# Reading electron micrographs

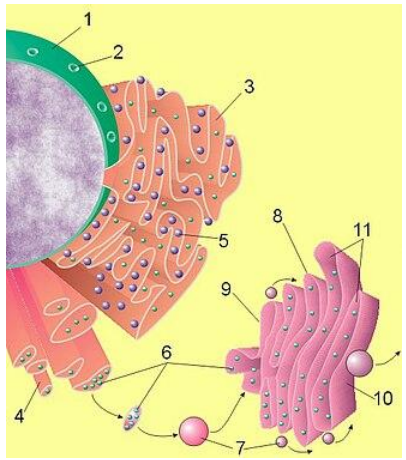
A2.2.10



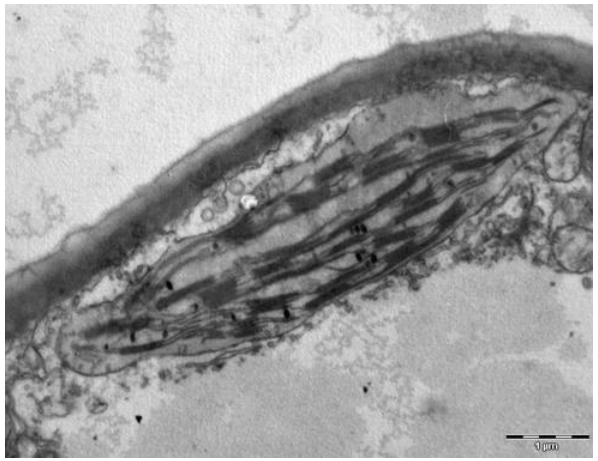

Mitochondria. Louisa Howard, public domain.



Microvilli. Louisa Howard, public domain.




Nucleus, ER, Golgi (diagram). Public domain.

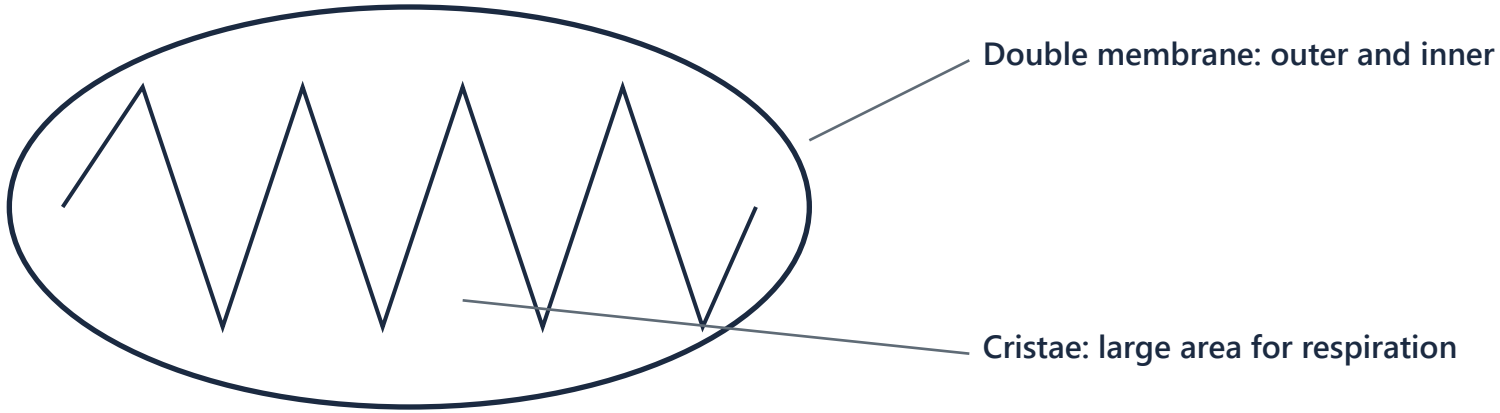


Chloroplast. CC BY-SA 3.0.

| Look for | It tells you |
|---|---|
| **Double membrane with folds inside** | Mitochondrion (cristae) |
| **Stacks of membranes in a lens shape** | Chloroplast (grana) |
| **Large round body with dark patches** | Nucleus (chromatin) |
| **Finger-like edge** | Microvilli: absorbing cell |

# Drawing from micrographs

A2.2.11




**Mitochondrion (x 30 000)**

1. Sharp pencil, single clear lines, no shading
2. Draw only what you see, in proportion
3. Ruled label lines that do not cross
4. Annotate: add what each part does
5. Title, and magnification or scale bar

# Practice: draw and annotate

A2.2.10–11



Mitochondria (TEM). Louisa Howard, public domain.

1. Draw one mitochondrion from this micrograph.
2. Label the outer membrane, inner membrane, cristae and matrix.
3. Annotate two of them with their function.
4. The mitochondrion is 1.5 μm long. Use the length of your drawing to state its magnification.

# Common mistakes to avoid

| Not this | But this |
| --- | --- |
| **Dividing mm by µm.** | Convert to the same unit first. |
| **Prokaryotes have no DNA.** | They have DNA, but no nucleus. |
| **Bacteria have no ribosomes.** | All cells have ribosomes; bacteria have 70S. |
| **All plant cells have chloroplasts.** | Root cells do not. |
| **Labelling with arrows and shading drawings.** | Ruled lines, no arrows, no shading. |

# Check that you can...

1. **Outline** cell theory
2. **Calculate** magnification and actual size, using scale bars
3. **Compare** light and electron microscopes, and **outline** newer techniques
4. **State** the structures common to all cells
5. **Compare** prokaryotic and eukaryotic cells
6. **Outline** how a single-celled organism carries out the functions of life
7. **Distinguish** animal, fungal and plant cells, and **describe** atypical cells
8. **Identify**, **draw** and **annotate** structures from micrographs

# Exit activity: 5 minutes

**On your own and without notes. Hand this in as you leave.**

1. A cell is 20 μm long and 40 mm long in an image. Calculate the magnification.
2. State three structures found in all cells.
3. Give one difference between the DNA of a prokaryote and a eukaryote.

# Image credits

Cork Micrographia Hooke.png – Robert Hooke – Public domain

Staphylococcus aureus 01.jpg – Photo Credit: Janice Carr Content Providers(s): CDC/ Matthew – Public domain

Paramecium.jpg – Barfooz at the English Wikipedia. – CC BY-SA 3.0

Red White Blood cells.jpg – Electron Microscopy Facility at The National Cancer Institut – Public domain

Compound Microscope (cropped).JPG – Acagastya – CC0

Mitochondria, mammalian lung – TEM.jpg – Louisa Howard – Public domain

FluorescentCells.jpg – Author not stated – Public domain

FreezefractureMU.jpg – Dr. Charles H. Mallery, Department of Biology, University of – Public domain

Average prokaryote cell- en.svg – Mariana Ruiz Villarreal, LadyofHats – Public domain

Animal cell structure en.svg – LadyofHats (Mariana Ruiz) – Public domain

Nucleus ER golgi.jpg – Author not stated – Public domain

Chloroplast in leaf of Anemone sp TEM 12000x.png – and3k and caper437 – CC BY-SA 3.0

Microvilli.jpg – Louisa Howard, Katherine Connollly – Dartmouth Electron Micr – Public domain

Paramecium bursaria – 400x (13263096305).jpg – Picturepest – CC BY 2.0

Plant cell structure-en.svg – LadyofHats – Public domain

Saccharomyces cerevisiae SEM.jpg – Mogana Das Murtey and Patchamuthu Ramasamy – CC BY 3.0

Skeletal muscle – longitudinal section.jpg – Author not stated – CC BY-SA 3.0

Hyphae.svg – SCIdude – CC0

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